

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\`a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-ad6e9c7e1bbd48711.jsonl`  
- SHA-256: `213c54bfcaa4d5dea04e2391547b4de0201cf3179979904959f9fe8328b4ec34`  
- Source modified: `2026-05-31T04:23:50+00:00`  
- Imported at: `2026-07-05T16:48:21+00:00`  
- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:23:13.554Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables):
TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"olmOCR-2-7B is a document-OCR-specialist VLM fine-tuned from Qwen2.5-VL-7B-Instruct, with additional GRPO RL training specifically to improve performance on math equations and tables ? the harder cases for bank-statement parsing."

****Source:**** <https://huggingface.co/allenai/olmOCR-2-7B-1025> (primary)

****Supporting quote:**** "This is a release of the olmOCR model that's fine tuned from Qwen2.5-VL-7B-Instruct using the olmOCR-mix-1025 dataset. It has been additionally fine tuned using GRPO RL training to boost its performance at math equations, tables, and other tricky OCR cases."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:23:15.999Z)

I'll verify this claim about olmOCR-2-7B. Let me check the primary source and search for any contradicting evidence.

3. assistant (2026-05-31T04:23:19.173Z)

I'll fetch the primary source to verify the specific claims about the fine-tuning lineage and GRPO RL training.

4. user (2026-05-31T04:23:25.094Z)

olmOCR-2-7B-1025 Model Details

Base Model

The base model is **Qwen2.5-VL-7B-Instruct**.

> "This is a release of the olmOCR model that's fine tuned from Qwen2.5-VL-7B-Instruct using the [olmOCR-mix-1025](https://huggingface.co/datasets/allenai/olmOCR-mix-1025) dataset."

Dataset

The primary dataset used is **olmOCR-mix-1025** for supervised fine-tuning (SFT).

GRPO RL Training

Yes, GRPO RL training is mentioned. Here's the exact quote:

> "It has been additionally fine tuned using GRPO RL training to boost its performance at math equations, tables, and other tricky OCR cases."

The specific capabilities targeted by GRPO RL training are:

- **Math equations**
- **Tables**
- **Other tricky OCR cases**

Additionally, there is a separate **olmOCR-synthmix-1025** RL dataset used for this reinforcement learning phase.

5. user (2026-05-31T04:23:31.712Z)

Web search results for query: "olmOCR-2-7B Qwen2.5-VL GRPO RL training math equations tables"

Links: [{"title": "olmOCR-2-7B ? Doubleword | Document OCR Model", "url": "https://www.doubleword.ai/models/olmocr-2-7b/"}, {"title": "[Literature Review] olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://www.themoonlight.io/en/review/olmocr-2-unit-test-rewards-for-document-ocr"}, {"title": "olmOCR 2: Unit Test Rewards for Document OCR", "url": "https://www.emergentmind.com/papers/2510.19817"}, {"title": "olmOCR 2: Unit test rewards for document OCR | Ai2", "url": "https://allenai.org/blog/olmocr-2"}, {"title": "olmOCR 2 Unit Test Rewards for Document OCR Jake Poznanski Luca Soldaini", "url": "https://arxiv.org/pdf/2510.19817"}, {"title": "allenai/olmOCR-2-7B-1025-FP8 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8"}, {"title": "allenai/olmOCR-2-7B-1025 · Hugging Face", "url": "https://huggingface.co/allenai/olmOCR-2-7B-1025"}, {"title": "Mungert/olmOCR-2-7B-1025-GGUF · Hugging Face", "url": "https://huggingface.co/Mungert/olmOCR-2-7B-1025-GGUF"}, {"title": "richardyoung/olmOCR-2-7B-1025-MLX-6bit · Hugging Face", "url": "https://huggingface.co/richardyoung/olmOCR-2-7B-1025-MLX-6bit"}, {"title": "olmOCR-2-7B-1025-FP8 huggingface.co api & allenai olmOCR-2-7B-1025-FP8 github AI Model - Toolify", "url": "https://www.toolify.ai/ai-model/allenai-olmocr-2-7b-1025-fp8"}]

Based on the search results, here's what I found about olmOCR-2-7B, Qwen2.5-VL, GRPO RL training, and their application to math equations and tables:

Overview

olmOCR-2-7B is fine-tuned from Qwen2.5-VL-7B with GRPO RL training for superior math, table, and document OCR. olmOCR 2 is built on Qwen2.5-VL-7B and fine-tuned on olmOCR-mix-1025, a dataset of 270,000 PDF pages with diverse properties?academic papers, historical scans, legal documents, brochures, and more.

Training Methodology

The olmOCR-2-7B-1025 VLM is initialized from a Qwen2.5-VL-7B-Instruct model fine-tuned on the olmOCR-mix-1025 dataset (267,962 pages), and is further trained using the Group Relative Policy

Optimization (GRPO) algorithm via the Hugging Face TRL library.

For each document, 28 completions are generated, and each completion is scored where each test case is either a pass or fail, and the reward is the fraction from 0.0 to 1.0 of passing test cases.

Performance on Math and Tables

RL training on test cases results in state-of-the-art performance on olmOCR-Bench, with the largest improvements in math formula conversion, table parsing, and multi-column layouts compared to previous versions.

Output Format

The model outputs natural-reading plain text with LaTeX for equations and HTML for tables.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-05-31T04:23:42.409Z)

Both the primary source and multiple independent secondary sources strongly corroborate the claim. Let me verify one element ? whether GRPO RL training is "additional" to the SFT lineage (i.e., the claim's framing that it's a Qwen2.5-VL fine-tune PLUS GRPO). The arXiv paper and Ai2 blog confirm the methodology precisely.

The claim has three sub-assertions, all verified:

1. *****fine-tuned from Qwen2.5-VL-7B-Instruct***** ? Confirmed verbatim by the model card and the arXiv paper ("initialized from a Qwen2.5-VL-7B-Instruct model").
2. *****additional GRPO RL training***** ? Confirmed. The arXiv paper (2510.19817) confirms it is "further trained using the Group Relative Policy Optimization (GRPO) algorithm via the Hugging Face TRL library" after SFT.
3. *****specifically to improve performance on math equations and tables***** ? Confirmed verbatim by the model card ("to boost its performance at math equations, tables, and other tricky OCR cases") and corroborated by the Ai2 blog/paper ("largest improvements in math formula conversion, table parsing, and multi-column layouts").

The source is a primary source (the official Ai2 HuggingFace model card), backed by a peer-style arXiv paper from the same authors. Not marketing fluff ? it's a technical model card with a corresponding paper. Current (1025 = Oct 2025 release). The claim is an accurate paraphrase, not an overreach.

One minor framing note: the claim says GRPO targets "the harder cases for bank-statement parsing" ? this is the *verifier's contextual gloss* connecting the model's stated table-improvement to the research question's use case, not something the source claims. But the core factual claim (lineage + GRPO + targets math/tables) is fully supported.

7. user (2026-05-31T04:23:47.633Z)

Structured output provided successfully

8. assistant (2026-05-31T04:23:50.488Z)

Claim verified as well-supported. The factual core ? olmOCR-2-7B is a Qwen2.5-VL-7B-Instruct fine-tune with additional GRPO RL training targeting math equations and tables ? is confirmed by the primary Ai2 model card and the companion arXiv paper (2510.19817). refuted=false, high confidence.